

Journal of Medical Economics

ISSN: 1369-6998 (Print) 1941-837X (Online) Journal homepage: www.tandfonline.com/journals/ijme20

Reforming the delivery of smoking cessation: a distributional cost-effectiveness analysis of providing smoking cessation as part of targeted lung cancer screening

H. Fewster, P. Nuttall, M. Evison, H. Holmes & R. Malcolm

To cite this article: H. Fewster, P. Nuttall, M. Evison, H. Holmes & R. Malcolm (2026) Reforming the delivery of smoking cessation: a distributional cost-effectiveness analysis of providing smoking cessation as part of targeted lung cancer screening, Journal of Medical Economics, 29:1, 1180-1194, DOI: 10.1080/13696998.2026.2655085

To link to this article: https://doi.org/10.1080/13696998.2026.2655085

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

View supplementary material

Published online: 13 Apr 2026.

Submit your article to this journal

Article views: 1427

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ijme20

JOURNAL OF MEDICAL ECONOMICS
2026, VOL. 29, NO. 1, 1180–1194
https://doi.org/10.1080/13696998.2026.2655085
Article /2655085

Taylor & Francis
by informa

ORIGINAL RESEARCH

OPEN ACCESS Check for updates

Reforming the delivery of smoking cessation: a distributional cost-effectiveness analysis of providing smoking cessation as part of targeted lung cancer screening

H. Fewster[a], P. Nuttall[a], M. Evison[a,b,c,d], H. Holmes[a] and R. Malcolm[a]

[a]York Health Economics Consortium, Heslington, UK; [b]Lung Cancer & Thoracic Surgery Directorate, Wythenshawe Hospital, Manchester University NHS Foundation Trust, Manchester, UK; [c]Manchester Academic Health Science Centre (MAHSC), Manchester, UK; [d]University of Manchester, Manchester, UK

ABSTRACT

Background: Lung cancer is a leading cause of cancer death, and smoking-related disease is a major cause of health inequality in England, driven by increased prevalence of smoking in deprived areas. Integrating smoking cessation support into the English Lung Cancer Screening (LCS) programme may optimize services and has been found cost-effective. However, limited evidence is available on its health equity impact.

Aim: The objective was to conduct an aggregate distributional cost-effectiveness analysis (DCEA) of providing smoking cessation as part of LCS compared with usual care (referral to stop smoking services**)** to estimate its impact on health inequalities and health-related social welfare in England.

Methods: The DCEA used an aggregate approach, stratifying people who smoke, aged 55–74 who were attending LCS by Index of Multiple Deprivation (IMD) quintiles. Discounted incremental costs and quality-adjusted life years (QALYs) were derived from a Markov model, adapted from previous NICE guidelines. The base case analysis applied a Health Opportunity Cost (HOC) of £15,000 per QALY and an Atkinson inequality aversion parameter (IAP) of 6.5, assuming a equal distribution across groups for intervention uptake.

Results: Under all base case assumptions, smoking cessation provided as part of targeted LCS had a positive net health benefit and health equity impact. The intervention dominated usual care, resulting in a total net health benefit of 142,035 QALYs. The distribution of benefit strongly favored the most deprived quintile (IMD1: 34,863 QALYs vs. IMD5: 23,612 QALYs), driven by smoking prevalence.

Conclusion: Reforming smoking cessation services as part of LCS would likely improve both population health and health equity. Uptake of smoking cessation is a key determinant of the impact on health inequalities, underscoring the importance of targeted implementation strategies for optimizing population health, while reducing health inequalities.

ARTICLE HISTORY

Received 15 January 2026
Revised 10 March 2026
Accepted 31 March 2026

KEYWORDS

Smoking cessation; lung cancer screening; health economics; distributional cost-effectiveness analysis; health inequalities

JEL CODES

I14; I15

PLAIN LANGUAGE SUMMARY

Lung cancer is one of the biggest causes of cancer deaths in England. Smoking is the leading cause of lung cancer and is more common in poorer communities, which increases health inequalities. Health inequalities are unfair, systematic, and avoidable differences in health status, life expectancy, and care access between population groups.

The English Lung Cancer Screening (LCS) programme invites people aged 55–74 who smoke or used to smoke for screening. Studies have already shown that offering stop-smoking support is good value for money and could improve health outcomes. However, it is not clear whether this would reduce or increase health inequalities.

This study examined what would happen if stop-smoking support was routinely provided as part of lung cancer screening, compared with usual care. The analysis

CONTACT R. Malcolm rob.malcolm@york.ac.uk York Health Economics Consortium, Enterprise House, Innovation Way, Heslington, YO10 5NQ, UK

Supplemental data for this article is available online at https://doi.org/10.1080/13696998.2026.2655085.

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.
www.tandfonline.com/ijme

grouped people by quantitative measures to capture health inequalities and estimated both costs and health benefits to different measures of deprivation.
The results showed that adding stop-smoking support to screening would improve overall health and reduce health inequalities. It would generate large health gains overall, with the greatest benefits seen in the most deprived groups, who have higher smoking rates. The programme would increase inequality if uptake of stop-smoking support was around 40% higher in less deprived groups than more deprived groups.
Overall, integrating stop-smoking services into lung cancer screening is likely to improve population health and narrow health gaps, especially if attempts are made to ensure strong uptake in more deprived areas.

Introduction

Despite a clear decline in smoking prevalence in recent decades, smoking remains an important contributor to preventable morbidity and mortality, particularly in high-risk and socioeconomically disadvantaged populations[1,2]. Notably, the lung cancer mortality rate in England was 63.5 and 44.4 for males and females respectively[3]. It is also a major risk factor for other serious diseases, such as cardiovascular disease (CVD), chronic obstructive pulmonary disease (COPD), pregnancy complications and infertility[4]. National mortality rates vary in part because smoking prevalence differs across regions and population groups[5] being disproportionately concentrated among people on lower incomes (i.e. a household income below 60% of the median UK household income)[6–8]. 2019 data in England revealed that individuals working in a managerial and professional role demonstrated notably lower prevalence rates than those in a routine and manual profession (10.2% and 25.4% respectively). It is also reinforced across generations through social norms, familiarization, and addiction. Early uptake of smoking exacerbates harm, as individuals who start young are more likely to become more dependent on smoking, have reduced likelihood of successful cessation, and experience higher mortality[9]. Reducing smoking rates in disadvantaged groups would therefore not only improve health outcomes but also support wider determinants of health, including through a reduction in poverty.

The English Lung Cancer Screening (LCS) programme offers a key opportunity to address the challenges regarding health inequality, and subsequent lung cancer risk. The programme invites people aged 55–74 who either smoke or used to smoke for a screening test. Integrating smoking cessation support directly into the LCS pathway may be an optimal strategy for delivering cessation services, potentially improving reach and efficacy compared with a central referral model.

Smoking cessation, as a public health intervention, is highly effective against preventing lung cancer and other associated chronic diseases. However, smoking cessation services are currently delivered through centralized services, where people can self-refer themselves, or be referred by clinicians. This separation results in high drop-off rates between referral and engagement, particularly among high-risk groups. Evidence indicates that delivering cessation support immediately at the point of screening is associated with a substantial increase in short term quit rates[10]. Long-term outcomes were not available in this study due to lack of follow-up data, however, evidence from the UKLS trial suggests that short-term quit rates are often reflective of long-term abstinence. Consequently, this emphasizes the importance of embedding smoking cessation within lung cancer screening programmes[11].

It should be noted that discrepancies between groups may be exacerbated if smoking cessation is excluded from healthcare screenings, particularly for those with higher socioeconomic status (SES). Excluding cessation support may benefit more advantaged participants, who can quit independently, whilst hindering lower SES groups who rely more heavily on embedded healthcare support rather than individuals close to them[6,12,13]. Individuals with poorer health literacy have indicated a lower likelihood of smoking cessation and higher likelihood of relapses[14,15]. Conversely, higher levels of health literacy are associated with significantly lower rates of smoking[16], suggesting a disproportionate advantage to those with greater SES. A study on bowel cancer screening identified that tailored educational programmes are required to reduce attrition rates in specific sub-populations[17]. For instance, a large proportion of participants within the discussion groups were of ethnic minority, therefore interpreters were

required. Subsequently, willingness to participate in educational talks increased from 41.5% to 85.7%. Providing an individualized approach could aid in increasing programme efficacy.

Data from the London SUMMIT cohort reported that 33.7% (2,090 out of 6,203) of people who currently smoke consented to a practitioner-made opt-out referral. Where outcome data were available, 47.3% ($n = 351/742$) accepted an appointment, of which 65.5% set a quit date. The overall 4-week validated quit rate was 17.8% among those with available outcome data[18]. Higher consent rates were seen among men and ethnic minority groups, though engagement was lower in the most deprived quintile, indicating both potential for improved reach and a need for targeted support. Integrating cessation services directly into screening appointments could therefore increase those taking up smoking cessation, leading to reduced smoking prevalence, and ultimately lower lung cancer incidence than central referral models.

A recent health economics study examined the cost-effectiveness of integrated stop smoking support in lung screening compared with usual care. Offering integrated smoking cessation support alongside LCS, was cost effective when compared with usual care, with a net monetary benefit of £2,198 per person[19]. However, the impact of this new model on existing health inequalities is currently unknown. The economic impact alone, assessed through standard cost-effectiveness analysis (CEA), does not account for the intervention's effects on health inequalities.

The correlation between socioeconomic status and poor health outcomes, particularly the increased risk of smoking-related disease, is well-established[6,20–23]. Data from the Health Survey for England shows that the prevalence of smoking is disproportionately concentrated in the most deprived Index of Multiple Deprivation (IMD) quintiles[2]. Therefore, understanding the impact of rolling smoking cessation into LCS is useful to understand the potential impact on health inequalities. IMD is a measure of accumulated social advantage and disadvantage of people in a particular area, such as deprivation in England[24]. This constitutes several components, including employment, education, health and income. Income is the most significant component, contributing to 22.5% of the total score[25]. However, a caveat of IMD is that as an ecological measure, it may not capture individual socioeconomic circumstances, highlighting an important limitation when interpreting inequality impacts[24].

DCEA is an economic evaluation tool that builds upon traditional CEA by assessing the equity impact and incorporating concerns about health inequality across population subgroups. This is achieved by estimating the distribution of incremental costs and health outcomes (Quality-Adjusted Life Years, QALYs) across these groups, providing a more comprehensive view of social welfare. This research involved conducting an aggregate DCEA of providing smoking cessation as part of LCS to estimate its combined effect on population health, health inequality, and health-related social welfare in England. This analysis follows the general methodological framework for DCEA developed for and implemented by the National Institute for Health and Care Excellence (NICE) in its evaluation processes[26].

Methods

An aggregate DCEA approach was applied to estimate the effect of smoking cessation provided as part of targeted LCS on population health and health equity in England. In aggregate DCEA, the incremental costs and benefits estimated in standard CEA are scaled to the size of the population and distributed across subgroups stratified by a measure of inequality, such as the IMD[27]. IMD is an index measure which captures key determinants of health, and are scored and weighted into one index, as a measure of deprivation. IMD includes the following domains: Income, Employment, Education/Skills/Training, Health/Disability, Crime, Barriers to Housing/Services, and the Living Environment[28].

As with all quantitative analysis, IMD may not reflect every form of deprivation or inequity that may arise but is a useful tool for quantitative analysis. A limitation of aggregate DCEA is that, unlike full DCEA, the cost-effectiveness results are assumed to be the same across all subgroups. However, conducting a full DCEA would require estimates stratified by a measure of inequality for a large number of parameters. This was not considered feasible, so an aggregate DCEA was chosen.

The cost-effectiveness results used in the aggregate DCEA were taken from a Markov model, which was adapted from previous NICE guidelines[19]. No changes to the underlying model itself were made regarding deprivation group. The initial NICE model aimed to assess the cost-effectiveness of a mass

media campaign, aiming at the prevention of the uptake of smoking among young people[29]. The Markov model was previously adapted to compare smoking cessation delivered alongside usual care to usual care (referral to stop smoking services).

The Markov model considered six long-term smoking-related illnesses: lung cancer, coronary heart disease, chronic obstructive pulmonary disorder, myocardial infarction, stroke and asthma. The prevalence of these illnesses broken down by age, sex and smoking status were captured using long-term epidemiological data. The impact on utility of the selected conditions and all-cause mortality were used to derive incremental costs and quality-adjusted life years (QALYs). The impact of providing smoking cessation as part of LCS was quantified by the estimated differences in the uptake of smoking cessation services, as well as different mixes of interventions (including no interventions), based on reported services offered and uptake.

The aggregate DCEA followed six stages. The first stage was to extract discounted incremental costs and QALYs from the Markov model[19]. In the second stage, the size and distribution of the base case population by socioeconomic status were estimated based on publicly available health data sets. The measure of socioeconomic status used was IMD, where IMD1 represents the most deprived quintile of the population and IMD5 represents the least deprived. The number of people taking up smoking cessation at LCS in each IMD quintile was estimated using a previously published estimates of uptake. This enabled the analysis of health outcomes for the intervention and usual care stratified by socioeconomic status.

In stages 3 and 4, the respective size of each IMD quintile and the cost-effectiveness results from the Markov model were used to calculate the population health benefit and health opportunity cost for each quintile[19]. This enabled the assessment of equity impacts and the derivation of social welfare changes. In stage 5, the net health benefits for each IMD quintile were calculated by combining the estimated distribution of health benefits and health opportunity costs. In stage 6, the resulting net health benefit distribution was combined with estimates of the baseline quality-adjusted life expectancy (QALE) derived from the literature to produce a post-intervention health distribution. The Atkinson welfare index was used to estimate the equity-weighted QALY impacts for the intervention and usual care[30].

Inputs

The inputs for the DCEA are summarized in Table 1. The general population size was sourced from the Office for National Statistics (ONS)[31]. This included adults in England aged 55–74 regardless of sex. The prevalence of individuals aged between 55 and 74 years who smoke by IMD quintile was sourced from Health Survey for England (HSE)[2]. HSE is an annual population survey which uses random sampling of private households in England to capture a representative view of health-related behaviors such as smoking, alcohol consumption, and general health status. As there was no data available on the prevalence of smoking by both age and IMD quintile, a risk ratio was calculated for individuals who currently smoke in the 55–74 age range compared to all ages, which was applied to the prevalence of people who currently smoke in all ages by IMD quintile. The uptake of smoking cessation at LCS was sourced from Murray et al.[32] As Murray et al. found no significant difference in uptake between IMD quintiles, it was assumed in the base case that uptake did not vary by IMD quintile.

The net benefit distribution across each IMD quintile was calculated using the clinical subpopulation size j (n_j) and the proportion of that subpopulation within each IMD quintile q (p_{qj}). These data were combined with the corresponding estimates of incremental QALYs (h_j) and costs (c_j) from the cost-effectiveness model. Using the following equation, the net health benefit for each socioeconomic group was then derived (b_q^p):

$$b_q^p = \sum_j \left(n_j h_j p_{qj} - \frac{n_j c_j}{k} o_q \right)$$

Here, k represents the health opportunity cost ratio (or cost-effectiveness threshold), and o_q denotes the proportion of health opportunity costs (HOC) falling to each socioeconomic group. The first term scales the per person health benefit to population-level and distributes based on the socioeconomic

Table 1. DCEA base case inputs.

Parameter	Value	Reference
Prevalence of individuals who currently smoke, by IMD		
IMD1	18.34%	Health Survey for England[2]
IMD2	14.92%	
IMD3	8.32%	
IMD4	9.30%	
IMD5	7.65%	
Health opportunity cost distribution by IMD		
IMD1	0.20	Anaya-Montes et al.[33]
IMD2	0.20	Assumed flat in base case.
IMD3	0.20	
IMD4	0.20	
IMD5	0.20	
Baseline QALE by IMD (S.E.)		
IMD1	62.19 (0.43)	Love-Koh et al.[34]
IMD2	65.54 (0.49)	Table 2: at birth
IMD3	69.54 (0.42)	
IMD4	71.12 (0.44)	
IMD5	73.27 (0.41)	
General population by IMD		
IMD1	2,054,762	Office for National Statistics 2020.[31]
IMD2	2,327,394	Calculated from all sex data for adults aged 55–74.
IMD3	2,634,936	
IMD4	2,762,784	
IMD5	2,777,147	
Uptake of smoking cessation at LCS by IMD		
IMD1	74.84%	NICE[13] equal distribution across groups assumed in base case.
IMD2	74.84%	Murray et al.[32]
IMD3	74.84%	
IMD4	74.84%	
IMD5	74.84%	

Abbreviations. IMD, Index of multiple deprivation; LCS, Lung cancer screening; QALE, Quality-Adjusted Life Expectancy.

Table 2. Incremental costs and QALYs base case results.

Intervention	Incremental costs	Incremental QALYs
Usual care	£33,750	8.89
Smoking cessation at LCS	£32,910	9.00

Abbreviations. LCS, Lung cancer screening; QALY, Quality-Adjusted Life Years.

Table 3. Equity-weighed QALYs and equity impact of smoking cessation offered at LCS compared to usual care by different levels of atkinson inequality aversion.

Inequality aversion	Implicit IMD1/IMD5 weight	Equity-weighted QALYs	Net equity impact
0 (no aversion)	1	142,035 (64,440, 219,629)	0 (0, 0)
1	1.19	144,510 (65,631, 223,389)	2,476 (1,115, 3,836)
2	1.41	147,083 (66,867, 227,300)	5,049 (2,272, 7,825)
3	1.67	149,744 (68,141, 231,347)	7,709 (3,466, 11,953)
4	1.99	152,480 (69,449, 235,511)	10,446 (4,692, 16,199)
5	2.37	155,278 (70,784, 239,772)	13,243 (5,944, 20,543)
6	2.81	158,122 (72,138, 244,106)	16,087 (7,215, 24,959)
6.5	**3.07**	**159,556 (72,820, 246,292)**	**17,521 (7,857, 27,186)**
7	3.35	160,996 (73,503, 248,488)	18,961 (8,500, 29,422)
8	3.99	163,882 (74,872, 252,892)	21,848 (9,791, 33,904)
9	4.75	166,764 (76,237, 257,291)	24,730 (11,082, 38,378)
10	5.67	169,625 (77,590, 261,659)	27,590 (12,364, 42,816)
10.95	**6.73**	**172,307 (78,857, 265,756)**	**30,272 (13,569, 46,975)**
11	6.76	172,447 (78,923, 265,970)	30,412 (13,632, 47,192)
12	8.07	175,214 (80,230, 270,199)	33,180 (14,879, 51,481)
13	9.63	177,913 (81,503, 274,323)	35,879 (16,098, 55,660)
14	11.5	180,531 (82,738, 278,324)	38,496 (17,284, 59,709)
15	13.73	183,056 (83,929, 282,182)	41,021 (18,432, 63,610)
16	16.41	185,479 (85,072, 285,885)	43,444 (19,539, 67,349)
17	19.62	187,793 (86,165, 289,420)	45,758 (20,600, 70,916)
18	23.45	189,992 (87,204, 292,780)	47,957 (21,613, 74,302)
19	28.05	192,074 (88,188, 295,960)	50,039 (22,577, 77,502)
20 (highest aversion)	33.55	194,036 (89,117, 298,956)	52,002 (23,489, 80,515)

Abbreviations. IMD, Index of multiple deprivation; QALYs, Quality-Adjusted Life Years.
Bold text represents literature estimates[37].
The table provides estimates for the equity-weighted QALYs and net equity impact and different levels of aversion to inequality. The results indicate that the net equity impact and equity weighted QALYs increase substantially as the aversion to inequalities increase. The implicit weight highlights how much more IMD1 QALYs are weighted than IMD5 at different aversion indices. For instance, a value of 10 for inequality aversion, weights the QALYs gained by people in IMD1 5.67 times greater than those gained in IMD5.

patterns estimated from HSE. The second term scales costs to population-level, converts them into health opportunity costs and distributes them across IMD quintiles based on existing distributional estimates. The HOC distribution by IMD quintile was assumed flat in the base case, following Anaya-Montes et al.[33] A HOC threshold of £15,000 per QALY was used, reflecting the Department of Health and Social Care's estimate of opportunity cost in the NHS.

To obtain the post-intervention QALE, the population net health benefit for each socioeconomic group was combined with estimates of "pre-intervention" quality-adjusted life expectancy (QALE).

$$H'_q = H_q + \frac{b_q^p}{n_q}$$

In this equation, H_q and H'_q represent the pre- and post-intervention QALE, respectively, while n_q is the total population in each IMD quintile. The baseline QALE by IMD quintile was sourced from Love-Koh et al.[34] The equity-weighted QALYs generated by the intervention were estimated using the Atkinson welfare index (Table 3). The Atkinson inequality aversion parameter (IAP) was set to 6.5 in the base case to align with Robson et al. (2024)[35]. This value implicitly weights health gains to the most deprived quintile (IMD1) approximately three times higher than in the least deprived quintile (IMD5).

Uncertainty

Parameter uncertainty was built into the analysis through probabilistic sensitivity analysis (PSA). It should be noted that this was applied to the DCEA only, and not to the underlying Markov model. Sensitivity and scenario analysis was used to explore how the health equity impact differed under alternative assumptions. The uncertainty in the HOC distribution was explored through two scenario analyses defined by Anaya Montes et al. in which the HOC slightly favored the most deprived and most affluent groups. A scenario analysis was also conducted using HOC estimates from Love-Koh et al.[36] Alternative HOC thresholds of £20,000 and £30,000 (to align with the lower and upper bounds used by NICE) were used to reflect the variety of thresholds used in healthcare decision making in England. Although, the alternative thresholds use do not represent HOC, so these results should be interpreted with caution.

Another area of uncertainty was the distribution of uptake of smoking cessation at LCS by IMD quintile. To assess the impact of a lower uptake among deprived groups, two scenario analyses were conducted in which the base case uptake was scaled up for the least deprived quintiles and scaled down for the most deprived groups by 25% and 50%. Sensitivity analyses were conducted using a range of Atkinson IAPs to assess the effect of inequality aversion on the health equity impact, including the 10.95 figure recommended by Robson et al. (2017).[37] Table S1 in the Supplementary Materials provides the scenario analysis inputs.

Results

The health benefit, health opportunity cost, and net benefit by IMD quintile for smoking cessation offered at LCS compared to usual care are shown in Figure 1. As a cost-effectiveness threshold value of £20,000 per QALY, health benefits and health opportunity costs totaled 96,671 and 45,364 QALYs respectively, yielding a total net health gain of 142,035 QALYs.

The positive health benefit indicates that providing smoking cessation as part of LCS is health improving compared to usual care. The distribution in health benefits effects reflects the prevalence of smoking and uptake of smoking cessation, with IMD1 receiving 27,790 (95% CI 11,433, 40,148) QALYs compared with 14,539 (95% CI 6,446, 22,633) for IMD5. The positive health opportunity cost indicates that providing smoking cessation as part of LCS is less expensive compared to usual care. This is distributed equally across quintiles due to the flat health opportunity curve used in the base case.

The positive net health benefit shows that overall population health would be increased as a result of providing smoking cessation as part of LCS, compared with usual care. IMD1 has the greatest net health benefit of 34,863 (16,099, 53,628) QALYs, followed by IMD2, and IMD4. IMD5 has the lowest net health net benefit, of 23,612 (10,387, 36,837) QALYs. This is driven by uptake smoking cessation by IMD and health opportunity cost of NHS spending.

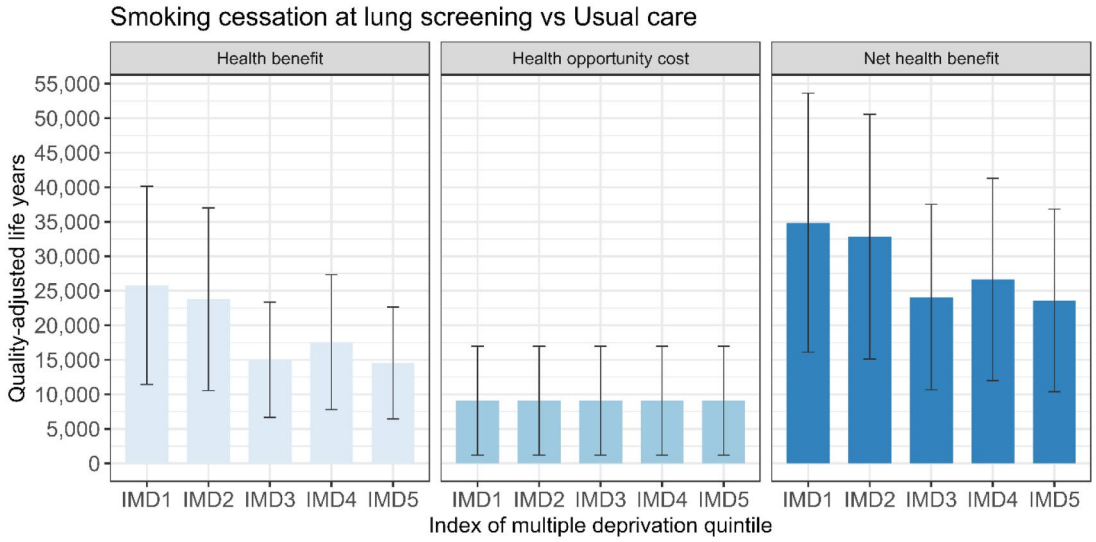


Figure 1. Health benefit, health opportunity cost and net health benefit by IMD quintile for smoking cessation offered at LCS compared to usual care.

The equity-weighted QALY impact was estimated using the Atkinson welfare index. The equity-weighted QALY was positive for all inequality aversion parameters and increased as inequality increased from 142,035 (95% CI 64,440, 219,629) QALYs for epsilon (ε) = 0 to 194,036 (95% CI 89,117, 298,956) QALYs for $\varepsilon = 20$. The corresponding equity impact was equivalent to a gain of 2,476 (154, 229) QALYs at the base case inequality aversion parameter of 1. At $\varepsilon = 6.5$, this was equivalent to a gain of 17,521 (7,857, 27,186). At $\varepsilon = 10.95$, this was equivalent to a gain of 30,272 (13,569, 46,975) QALYs. These results are shown in Table 4.

The results from each of the PSA iterations are plotted on the equity-efficiency impact plane shown in Figure 2. The base case Atkinson inequality aversion parameter of $\varepsilon = 6.5$ was used. This graph indicates that smoking cessation as part of LCS has a positive net health benefit, and a positive equity impact compared with usual care, under base case assumptions. The PSA cloud is close to the y-axis, indicating that the results are sensitive to key data points such as uptake and prevalence

Scenario analyses

Table 4 shows the health benefit, health opportunity cost, and net health benefit by IMD quintile for the scenario and sensitivity analyses. The Supplementary Materials includes equity-impact planes for the scenario and sensitivity analyses.

One scenario was conducted using an alternative opportunity cost threshold: £30,000 per QALY. The overall health opportunity cost of providing smoking cessation as part of LCS decreased to 22,838 QALYs when the £30,000 threshold was used. Here, smoking cessation as part of LCS was health and equity improving, with an overall net health benefit of 120,235 QALYs. The highest net health benefit was in IMD1 at 35,933 (15,716, 56,150) QALYs and the lowest was in IMD5 at 22,798 (9,591, 36,004).

Two scenarios were conducted to assess the impact of different health opportunity costs distributions on the DCEA. When a pro-deprived health opportunity cost distribution was used, the net health benefit increased in the more deprived IMD quintiles compared to the base case, with a net health benefit of 35,933 (15,716, 56,150) QALYs in IMD1 compared to 22,798 (9,591, 36,004) QALYs in IMD5. This reflects the positive health opportunity cost of LCS as part of usual care, as the more deprived quintiles experienced a greater gain from the cost savings from the intervention. The equity impact increased for the most deprived quintiles but slowly decreased as deprivation decreased. Conversely, when a pro-affluent health opportunity cost distribution was applied, the net health benefit increased in less deprived IMD quintiles compared to more deprived quintiles.

Two scenarios were conducted to assess the impact of uptake. When a distribution in uptake of 25% was applied, IMD4 received the highest health benefit of 32,009 (14,494, 49,524) QALYs. Conversely,

Table 4. Health benefit, health opportunity cost and net health benefit by IMD quintile for smoking cessation offered at LCS compared to usual care in scenario analyses.

Outcomes (in QALYs)	IMD1 (most deprived)	IMD2	IMD3	IMD4	IMD5 (least deprived)	Total
Base case						
Health benefit	25,790 (11,433, 40,148)	23,763 (10,534, 36,992)	15,008 (6,654, 23,362)	17,570 (7,790, 27,351)	14,539 (6,446, 22,633)	96,671
Health opportunity cost	9,073 (1,188, 16,957)	9,073 (1,188, 16,957)	9,073 (1,188, 16,957)	9,073 (1,188, 16,957)	9,073 (1,188, 16,957)	45,364
Net health benefit	34,863 (16,099, 53,628)	32,836 (15,104, 50,568)	24,081 (10,637, 37,524)	26,643 (11,982, 41,304)	23,612 (10,387, 36,837)	142,035
Cost-effectiveness threshold: £20,000 per QALY						
Health benefit	25,601 (11,518, 39,684)	23,589 (10,615, 36,562)	14,899 (6,703, 23,095)	17,443 (7,849, 27,037)	14,433 (6,492, 22,375)	95,964
Health opportunity cost	6,835 (789, 12,881)	6,835 (789, 12,881)	6,835 (789, 12,881)	6,835 (789, 12,881)	6,835 (789, 12,881)	34,177
Net health benefit	32,436 (15,126, 49,746)	30,424 (14,159, 46,689)	21,734 (9,842, 33,626)	24,278 (11,136, 37,420)	21,269 (9,600, 32,937)	130,141
Cost-effectiveness threshold: £30,000 per QALY						
Health benefit	25,984 (11,708, 40,260)	23,941 (10,787, 37,095)	15,121 (6,815, 23,427)	17,702 (7,979, 27,426)	14,649 (6,601, 22,697)	97,397
Health opportunity cost	4,568 (530, 8,605)	4,568 (530, 8,605)	4,568 (530, 8,605)	4,568 (530, 8,605)	4,568 (530, 8,605)	22,838
Net health benefit	30,551 (14,317, 46,786)	28,508 (13,364, 43,652)	19,689 (9,174, 30,203)	22,270 (10,420, 34,120)	19,217 (8,943, 29,491)	120,235
Pro-deprived HOC distribution (Anaya-Montes et al.)						
Health benefit	25,952 (11,244, 40,660)	23,912 (10,360, 37,465)	15,103 (6,541, 23,664)	17,682 (7,660, 27,704)	14,631 (6,337, 22,925)	97,280
Health opportunity cost	9,981 (993, 18,969)	9,527 (947, 18,107)	9,074 (902, 17,245)	8,620 (857, 16,383)	8,166 (812, 15,520)	45,368
Net health benefit	35,933 (15,716, 56,150)	33,440 (14,586, 52,293)	24,176 (10,064, 38,288)	26,302 (11,255, 41,349)	22,798 (9,591, 36,004)	142,649
Pro-affluent HOC distribution (Anaya-Montes et al.)						
Health benefit	25,683 (10,770, 40,596)	23,666 (9,924, 37,408)	14,947 (6,267, 23,627)	17,499 (7,336, 27,661)	14,481 (6,073, 22,890)	96,276
Health opportunity cost	8,122 (985, 15,259)	8,573 (1,040, 16,107)	9,025 (1,094, 16,955)	9,476 (1,149, 17,802)	9,927 (1,204, 18,650)	45,123
Net health benefit	33,805 (14,817, 52,793)	32,239 (14,061, 50,417)	23,971 (10,020, 37,922)	26,975 (11,420, 42,529)	24,408 (10,024, 38,792)	141,398
Love-Koh et al. HOC distribution						
Health benefit	25,946 (11,588, 40,305)	23,908 (10,679, 37,138)	15,100 (6,746, 23,454)	17,677 (7,897, 27,458)	14,628 (6,534, 22,721)	97,260
Health opportunity cost	11,962 (1,309, 22,615)	9,951 (1,089, 18,814)	9,892 (1,083, 18,702)	7,274 (796, 13,752)	6,299 (689, 11,908)	45,378
Net health benefit	37,908 (16,987, 58,829)	33,860 (15,312, 52,408)	24,993 (10,704, 39,282)	24,951 (11,295, 38,608)	20,926 (9,437, 32,415)	142,638
Slight uptake distribution						
Health benefit	17,491 (7,588, 27,393)	16,115 (6,993, 25,238)	15,093 (6,546, 23,639)	23,423 (10,162, 36,685)	19,383 (8,411, 30,355)	91,505
Health opportunity cost	8,586 (1,112, 16,059)	8,586 (1,112, 16,059)	8,586 (1,112, 16,059)	8,586 (1,112, 16,059)	8,586 (1,112, 16,059)	42,929
Net health benefit	26,077 (11,586, 40,567)	24,701 (10,892, 38,510)	23,679 (10,364, 36,993)	32,009 (14,494, 49,524)	27,969 (12,530, 43,407)	134,434
Moderate uptake distribution						
Health benefit	8,994 (4,023, 13,964)	8,287 (3,708, 12,866)	15,007 (6,713, 23,302)	25,265 (11,300, 39,231)	20,907 (9,353, 32,460)	78460
Health opportunity cost	7,444 (992, 13,896)	7,444 (992, 13,896)	7,444 (992, 13,896)	7,444 (992, 13,896)	7,444 (992, 13,896)	37220
Net health benefit	16,438 (6,847, 26,028)	15,731 (6,445, 25,017)	22,451 (10,074, 34,829)	32,709 (15,140, 50,278)	28,351 (13,031, 43,671)	115680

Abbreviations. IMD, Index of multiple deprivation; HOC, Health opportunity cost; QALY, Quality-Adjusted Life Years.
*Confidence interval in parentheses.
**Health opportunity costs are calculated using a cost-effectiveness threshold of £15,000 per quality-adjusted life year and distributed using a flat health opportunity cost unless otherwise stated.

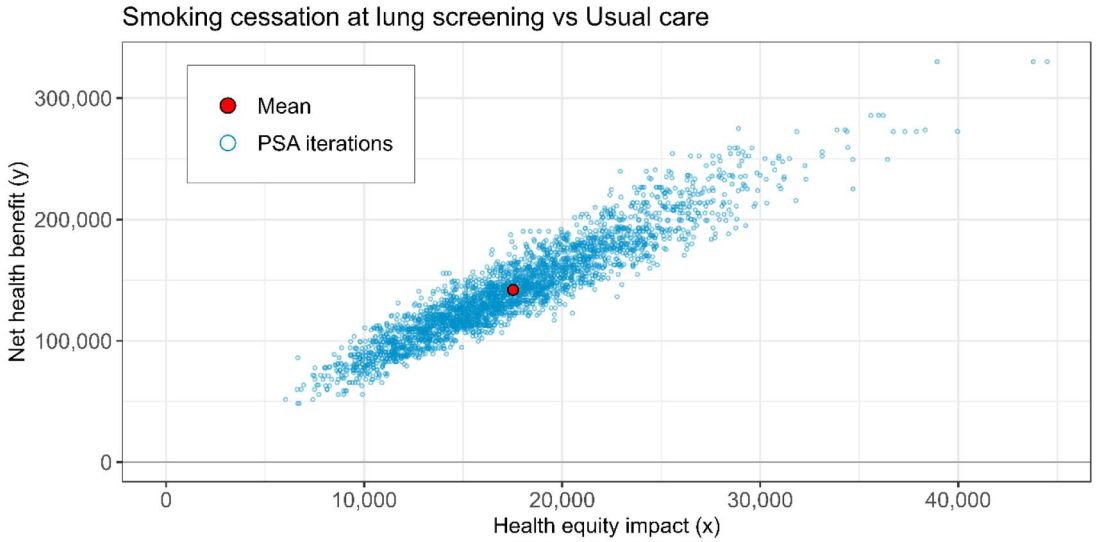


Figure 2. Equity-impact plane for smoking cessation offered at LCS compared to usual care.

IMD3 had the smallest net health benefit of 23,679 (10,364, 36,993) QALYs. When a moderate uptake distribution was applied, IMD2 had the smallest net health benefit of 15,731 (6,445, 25,017) QALYs, followed by IMD1 (16,438 (6,847, 26,028) QALYs). Again, IMD4 had the greatest net health benefit of 32,709 (15,140, 50,278) QALYs.

Threshold analysis was conducted to determine the uptake distribution where the results would change direction. The crossover point from a positive to negative equity impact occurred approximately between a 35% and 40% pro-affluent uptake distribution. The results were also considered for the cross over point of prevalence and suggest that the prevalence would have to be close to a 30% shift from current prevalence estimates towards the least deprived, to change the direction of the results.

Discussion

In accordance with all base case assumptions, the results of this aggregate DCEA suggest that smoking cessation, provided as part of targeted LCS, has a positive net health benefit and equity impact, compared with usual care. The intervention dominated usual care, resulting in a total net health benefit of 142,035 QALYs. Moreover, individuals within the most deprived quintile received the greatest benefit, compared to the least deprived (IMD1: 34,863 QALYs vs. IMD5: 23,612 QALYs respectively). This distribution of benefit was driven by smoking prevalence, in accordance with previous estimates from the Health Survey for England[2].

Additional scenario analyses were conducted to explore how the health equity impact differed under alternative assumptions. When a pro-deprived health opportunity cost distribution was used, the NHB increased in the more deprived IMD quintiles compared to the base case, with a net health benefit of 35,933 (15,716, 56,150) QALYs in IMD1 compared to 22,798 (9,591, 36,004) QALYs in IMD5. Conversely, when a pro-affluent health opportunity cost distribution was applied, the NHB increased in less deprived IMD quintiles compared to more deprived quintiles. The crossover point from a positive to negative equity impact occurred approximately between $a \pm 35\%$ and $\pm 40\%$ pro-affluent uptake distribution. Therefore, uptake would have to be substantially higher in the least deprived, for the policy to not improve health inequalities, and the distribution of health opportunity costs does not change the direction of the results. It is unlikely that uptake would be over 35% greater in the least deprived than the most deprived, given that attendance at lung cancer screening is already higher from people in more deprived areas[32].

Evidence suggests that smoking cessation is a cost-effective intervention, and is widely reported throughout literature, regardless of the chosen method of intervention[38]. Previous literature has shown that combination pharmacotherapy and behavioral support is the most effective, and cost-effective approach.

The analysis found that highly affluent individuals who smoke display a higher probability of utilizing services and greater quit rates among receiving them. Though despite this, all interventions were expected to reduce health inequalities. This was due to higher prevalence of smokers in lower socioeconomic groups, further exacerbated by the reduction in health inequality from cost savings.

The method of DCEA is focusing on deprivation, measured by IMD. However, we acknowledge that health inequalities are complex, intersectional, with lots of factors that cause and impact them. The purpose of this paper is not to suggest that health inequalities will be perfectly solved by improving smoking cessation services. The purpose of this paper is to understand how optimizing smoking cessation services may impact health inequalities. We acknowledge that in the longer-term, a wide range of policies are required to substantially reduce health inequalities, well beyond just healthcare.

However, evidence shows that effective smoking cessation support can reduce health inequalities, most notably when services are designed to target those at the greatest risk[39,40]. In line with the present findings, prior DCEA work on smoking cessation has highlighted that both smoking rates and baseline health risk are concentrated in more deprived groups. Smoking cessation interventions could subsequently generate substantial health gains in these groups and reduce overall inequality[41]. However, the current design of smoking cessation services can hinder proportionality of service utilization. In other words, traditional services tend to be disproportionately used by more affluent groups[39,41,42]. It is therefore vitally important that future research continues to engage with target populations, and understand the barriers to accessing services, to maximize the potential impact of optimizing smoking cessation service.

Embedding cessation support within LCS based on the evidence presented in this paper is likely to impact those in more deprived areas more. Between 2013 and 2017, lung cancer incidence rates in England were 174% higher in the most deprived quintile compare with the least, and males were 168% higher in the most deprived quintile compared with the least[43]. In theory, placing an intensive intervention within this pathway should reduce rates of unequal access to care. The DCEA findings support this argument, in which the NHB is largest in IMD1, and equity-weighted QALYs remain positive across all inequality aversion parameters. The threshold analysis indicates that inequalities would only widen under an uptake pattern where the least deprived have a substantially higher uptake, which is unlikely. In practice, uptake of LCS screening is already higher in more deprived populations, implying that the process of delivering cessation through LCS, supports equity gains rather than equity losses. Moreover, this helps explain why no substantial skew between quintiles is observed in the current findings (Figure 1). Therefore, embedding smoking cessation services within lung cancer screening is likely to be cost-effective, but also likely to improve health inequalities.

However, highly deprived people who smoke often face substantial private costs when engaging with cessation interventions. These include travel expenses, time constraints, and the ongoing cost of therapy following the initial referral. Such costs can limit uptake and sustained engagement within interventions[44]. Limited health literacy or confidence surrounding cessation programmes may further reduce continuation and adherence. Moreover, if the opportunity cost of engaging with the intervention is higher for more deprived groups, for example due to unstable or unflexible employment, the net benefit they receive may be lower than estimated. These considerations imply that service redesign should incorporate active efforts to minimize the financial burden for high-risk groups.

Strengths

The cost-effectiveness results used in the aggregate DCEA were taken from a Markov model, adapted from previous NICE guidelines[19]. The model considered six long-term smoking-related illnesses, in which each illness was broken down by age, sex, and smoking status using long-term epidemiological data. The Markov model was adapted from a previous model, with adjustments made to the cessation intervention groups. Moreover, model validation within published literature enhances the validity and generalizability of the model.

Previous Markov models, including the BENESCO model, identified comparable results with the current model. The BENESCO model is used to estimate the lifetime cost and benefits of smoking, and has been applied across a variety of populations, including the USA, the Caribbean, central America,

and Europe[45–47]. In accordance with the adapted NICE model, smoking cessation was found more cost-effective than behavioural support and reduced the likelihood of smoking-related diseases later in life. Substantial health benefits and cost savings were notable effects of the intervention. Therefore, to have a robust aggregate DCEA, a robust cost-effectiveness model is required.

A major strength of the model is the extensive availability of high-quality, published data on smoking behavior in England. Key model inputs are more certain than typical for a DCEA. The general population size was sourced from the ONS[31], including adults between 55 and 74 years. Moreover, prevalence rates were extracted from the HSE[2]. Both sources hold strong internal validity for population-level smoking estimates, including smoking prevalence, cessation rates, and socioeconomic distribution. In addition, the model draws upon well-established epidemiological parameters and uses nationally representative distributions rather than assumptions or imputed values, which substantially reduces structural uncertainty.

Limitations

No data was available on the prevalence of smoking by both age and IMD quintile, therefore a risk ratio was calculated for people who smoke in the 55 to 74 age range, compared to all ages. This approach introduces model uncertainty, as it assumes that the relative difference in smoking rates is proportional across all deprivation quintiles. Relying on a risk-ratio adjustment rather than empirical estimates could misrepresent the true socio-economic distribution in smoking for the screening-eligible age group.

Moreover, the analysis does not consider all potential policies related to cancer screening within the UK. Further analysis could investigate the impact of drug treatment in conjunction with the current strategy. It is evident in literature that highly deprived individuals display more severe rates of tobacco addiction, particularly from a younger age[9,48,49]. World Bank studies of household disposable income revealed that for the poorest households with at least one smoker, approximately 10% of disposable income was used for tobacco[50]. Therefore, combining drug treatment, such as nicotine replacement therapy (NRT) with smoking cessation as part of lung cancer screening could maximize the health outcomes for vulnerable individuals. Additionally, tailored cancer screening programmes have shown promising findings regarding participation and developing health literacy, particularly for those of ethnic minorities[17]. Future research could look to consider these alternative strategies to improve screening uptake.

Conducting a full DCEA was not feasible for the current analysis, therefore aggregate DCEA was chosen. Unlike full DCEA, aggregate DCEA assumes the cost-effectiveness results to be the same across all subgroups. This is unlikely to hold true for smoking-related interventions, as smoking cessation success[51], disease risk[52], and engagement with services may all vary systematically according to socioeconomic status[18]. A substantial proportion of literature indicates that individuals in more deprived quintiles experience greater difficulty in quitting, suffer with higher baseline morbidity, and engage differently with screening and cessation services[9,51,52].

Nevertheless, the main findings still suggest that the policy change is likely to be beneficial for health inequalities, as the observed distributional effects were large and consistent across multiple scenario analyses, with all reasonable variations in effectiveness falling well within the same inequality distribution. Consequently, while the assumption of equal effectiveness may affect absolute estimates, it is unlikely to overturn the conclusions of the study.

Finally, it is also important to consider the limitations associated with DCEA more broadly This method is focused on stratifying outcomes by a measure of deprivation, in this case, IMD. This is a metric which quantifies a range of factors impacting deprivation into one index measure. However, it is important to consider its limitations, with respect to how the integration of smoking cessation services could impact on more specific groups in society. As outlined in previous literature, there is large intersectionality between characteristics, which can lead to inequalities, and are likely to be simplified within quantitative analysis[53]. We fully support that further research could explore this by looking at other specific characteristics which are associated with health inequalities.

Conclusion

The aggregate DCEA framework outlined in this paper demonstrates how concerns for health inequality can be taken into account when evaluating health care interventions funded within a fixed health budget. The objective of this work was to estimate the expected joint effects of redesigning smoking cessation services to be embedded within LCS on population health and health inequality in England. An aggregate DCEA approach was taken. Linked, nationally representative health survey data in England was used to estimate the socioeconomic distribution.

The results found that there is a positive impact on population health and health equity, compared to current centralized smoking cessation services. This is due to those attending LCS being more likely to be more deprived, and higher engagement with smoking cessation results in improving population health and reducing health inequalities. The scenario analyses that were conducted show that the results are not sensitive to assumptions on health opportunity costs. Only if uptake was substantially higher in the least deprived (by over 35%), does the impact on health inequalities change. This is unlikely, given those who attend screening already tend to be from more deprived areas.

The results highlight the importance of targeted implementation plans in maximizing total health benefit and reducing inequalities for smoking cessation. Therefore, considering more active and decentralized services may be beneficial for reducing smoking harms, and reducing health inequalities in England.

Transparency

Declaration of funding

No funding was received for this manuscript.

Declaration of financial/other interests

No conflict of interests to declare. Peer reviewers on this manuscript have no relevant financial or other relationships to disclose.

Author contributions

The analysis was developed by HF and RM, extending previous work by RM, HH and ME. ME provided data and input for the analysis. The first draft of the manuscript was developed by HF and PN. RM, ME and HH reviewed 2 rounds of the manuscript and co-drafted the discussion. The manuscript was then finalized by all co-authors.

Acknowledgements

None stated.

Ethics statement

This article is a secondary data analysis using previously published and/or publicly available data, therefore it does not require ethical approval (or IRB review/oversight) as it does not involve the collection of new data from human participants. All data used were de-identified and obtained from publicly accessible sources, and the original studies obtained all necessary informed consents and ethical approvals.

ORCID

H. Fewster http://orcid.org/0009-0009-4721-1538
P. Nuttall http://orcid.org/0009-0007-5449-1414
M. Evison http://orcid.org/0000-0001-5148-3928
H. Holmes http://orcid.org/0000-0003-2559-061X

References

[1] Action on Smoking and Health. Smoking Statistics 2025. Available from: https://ash.org.uk/uploads/Smoking-Statistics-Fact-Sheet-2025.pdf?v=1764162649.

[2] NHS England. Health Survey for England 2022. Part 1: Data Tables; 2024.

[3] UK G. Cancer registration statistics: cancer mortality in England, 2018; 2020. Available from: https://www.gov.uk/government/statistics/cancer-registration-statistics-cancer-mortality-in-england-2018/cancer-registration-statistics-cancer-mortality-in-england-2018.

[4] Excellence NIfHaC. Smoking cessation: what are the harms caused by smoking?; 2025. Available from: https://cks.nice.org.uk/topics/smoking-cessation/background-information/harms-caused-by-smoking/.

[5] Jackson SE, Brown J, Buss V, et al. Sociodemographic and regional differences in cigarette consumption across Great Britain: a population study, 2022-2024. Nicotine Tob Res. 2025;27(12):2289–2296. doi: 10.1093/ntr/ntaf133.

[6] Department of Health and Social Care. Smoking and Inequalities 2024. Available from: https://fingertips.phe.org.uk/profile/tobacco-control/supporting-information/smokingandinequalities.

[7] Parliment U. Poverty in the UK: statistics 2025. Available from: https://commonslibrary.parliament.uk/research-briefings/sn07096/#:~:text=How%20is%20poverty%20measured?,absolute%20poverty)%20after%20housing%20costs.

[8] Jha P, Peto R, Zatonski W, et al. Social inequalities in male mortality, and in male mortality from smoking: indirect estimation from national death rates in England and Wales, Poland, and North America. Lancet. 2006;368(9533):367–370. doi: 10.1016/S0140-6736(06)68975-7.

[9] AoSa H. Young people and smoking 2025. Available from: https://ash.org.uk/resources/view/young-people-and-smoking.

[10] Buttery SC, Williams P, Mweseli R, et al. Immediate smoking cessation support versus usual care in smokers attending a targeted lung health check: the QuLIT trial. BMJ Open Respir Res. 2022;9(1):e001030. doi: 10.1136/bmjresp-2021-001030.

[11] Anthenelli RM, Benowitz NL, West R, et al. Neuropsychiatric safety and efficacy of varenicline, bupropion, and nicotine patch in smokers with and without psychiatric disorders (EAGLES): a double-blind, randomised, placebo-controlled clinical trial. Lancet. 2016;387(10037):2507–2520. doi: 10.1016/S0140-6736(16)30272-0.

[12] Chandola T, Head J, Bartley M. Socio-demographic predictors of quitting smoking: how important are household factors? Addiction. 2004;99(6):770–777. doi: 10.1111/j.1360-0443.2004.00756.x.

[13] Hiscock R, Dobbie F, Bauld L. Smoking cessation and socioeconomic status: an update of existing evidence from a national evaluation of English stop smoking services. Biomed Res Int. 2015;2015:274056–274010. doi: 10.1155/2015/274056.

[14] Li M, Sonoda N, Koh C, et al. Meta-analysis of the association between health literacy and smoking. Popul. Med. 2022;4(August):1–11. doi: 10.18332/popmed/152572.

[15] National Centre for Smoking Cessation and Training (NCSCT). Stop smoking services and health inequalities 2025. Available from: https://www.ncsct.co.uk/library/view/pdf/stop-smoking-services-and-health-inequalities.pdf.

[16] Tao Z, Xu Q, Zhu Y, et al. The impact of health literacy on smoking patterns among male residents: insights from Ningbo City. Front Public Health. 2025;13:1487400. doi: 10.3389/fpubh.2025.1487400.

[17] Tai CK, Leung P, Poullis A, et al. Comment on 'Reasons for non-uptake and subsequent participation in the NHS bowel cancer screening programme: a qualitative study. Br J Cancer. 2015;112(11):1834–1834. doi: 10.1038/bjc.2014.559.

[18] Bhamani A, Katsampouris E, Bojang F, et al. Uptake and 4-week outcomes of an 'opt-out' smoking cessation referral strategy in a London-based lung cancer screening setting. BMJ Open Respir Res. 2025;12(1):e002337. doi: 10.1136/bmjresp-2024-002337.

[19] Evison M, Naylor R, Malcolm R,et al. Health economic model to evaluate the cost-effectiveness of smoking cessation services integrated within lung cancer screening in the United Kingdom. Thorax BMJ. 2025. doi: 10.1136/thorax-2024-222815.

[20] McLean G, Gunn J, Wyke S, et al. The influence of socioeconomic deprivation on multimorbidity at different ages: a cross-sectional study. Br J Gen Pract. 2014;64(624):e440–e447. doi: 10.3399/bjgp14X680545.

[21] Goodman E, Slap GB, Huang B. The public health impact of socioeconomic status on adolescent depression and obesity. Am J Public Health. 2003;93(11):1844–1850. doi: 10.2105/ajph.93.11.1844.

[22] Arpey NC, Gaglioti AH, Rosenbaum ME. How socioeconomic status affects patient perceptions of health care: a qualitative study. J Prim Care Community Health. 2017;8(3):169–175. doi: 10.1177/2150131917697439.

[23] Winkleby MA, Jatulis DE, Frank E, et al. Socioeconomic status and health: how education, income, and occupation contribute to risk factors for cardiovascular disease. Am J Public Health. 1992;82(6):816–820. doi: 10.2105/ajph.82.6.816.

[24] thebmj. Individual socioeconomic status is important but hard to measure 2021. Available from: https://www.bmj.com/content/372/bmj.n174#:~:text=But%20it%20is%20important%20to,up%20with%20a%20core%20list.

[25] UK G. Indices of Deprivation 2019: income and employment domains combined for England and Wales – guidance note 2020. Available from: https://www.gov.uk/government/statistics/indices-of-deprivation-2019-income-and-employment-domains-combined-for-england-and-wales/indices-of-deprivation-2019-income-and-employment-domains-combined-for-england-and-wales-guidance-note#:~:text=The%20Income%20Deprivation%20Domain%20measures,in%20the%20IoD2019%20technical%20report.
[26] National Institute for Health and Care Excellence (NICE). Support document: health inequalities 2025. Available from: https://www.nice.org.uk/process/pmg36/resources/support-document-health-inequalities-pdf-19845921027013.
[27] Asaria M, Griffin S, Cookson R. Distributional cost-effectiveness analysis: a tutorial. Med Decis Making. 2016; 36(1):8–19. doi: 10.1177/0272989X15583266.
[28] UK G. Accredited official statistics. English indices of deprivation 2025. Statistical release 2025 Available from: https://www.gov.uk/government/statistics/english-indices-of-deprivation-2025/english-indices-of-deprivation-2025-statistical-release.
[29] National Institute for Health and Care Excellence (NICE). Tobacco: preventing uptake, promoting quitting and treating dependence. PH14 Economic Modelling Report. 2021. Available from: https://www.nice.org.uk/guidance/ng209/evidence/ph10-economic-modelling-report-pdf-392073181525
[30] Atkinson AB. On the measurement of inequality. J Econ Theory. 1970;2(3):244–263. doi: 10.1016/0022-0531(70)90039-6.
[31] Office for National Statistics. Population by index of multiple deprivation (IMD), England, 2001 to 2019 newport: office for national statistics. 2020. Available from: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/adhocs/12386populationbyindexofmultipledeprivationimdengland2001to2019.
[32] Murray RL, Alexandris P, Baldwin D, et al. Uptake and 4-week quit rates from an opt-out co-located smoking cessation service delivered alongside community-based low-dose computed tomography screening within the Yorkshire Lung Screening Trial. Eur Respir J. 2024;63(4):2301768. doi: 10.1183/13993003.01768-2023.
[33] Anaya-Montes M, Grašič K, Lomas J, et al. CHE Research Paper 197: do the poor gain more? The impact on health inequality of changes in public expenditure on secondary care. York2025. Available from: https://www.york.ac.uk/media/che/documents/papers/researchpapers/CHE%20RP%20197.pdf
[34] Love-Koh J, Schneider P, McNamara S, et al. Decomposition of quality-adjusted life expectancy inequalities by mortality and health-related quality of life dimensions. Pharmacoeconomics. 2023;41(7):831–841. doi: 10.1007/s40273-023-01264-9.
[35] Robson M, O'Donnell O, Van Ourti T. Aversion to health inequality—pure, income-related and income-caused. J Health Econ. 2024;94:102856. doi: 10.1016/j.jhealeco.2024.102856.
[36] Love-Koh J, Cookson R, Claxton K, et al. Estimating social variation in the health effects of changes in health care expenditure. Med Decis Making. 2020;40(2):170–182. doi: 10.1177/0272989X20904360.
[37] Robson M, Asaria M, Cookson R, et al. Eliciting the level of health inequality aversion in England. Health Econ. 2017;26(10):1328–1334. doi: 10.1002/hec.3430.
[38] Braun A, Hyll W, Krczal E. Breaking the habit: a systematic review of the cost-effectiveness of non-pharmacological and combined interventions for smoking cessation in Europe. Eur J Health Econ. 2025; doi: 10.1007/s10198-025-01855-7.
[39] Hiscock R, Murray S, Brose LS, et al. Behavioural therapy for smoking cessation: the effectiveness of different intervention types for disadvantaged and affluent smokers. Addict Behav. 2013;38(11):2787–2796. doi: 10.1016/j.addbeh.2013.07.010.
[40] Kock L, Brown J, Hiscock R, et al. Individual-level behavioural smoking cessation interventions tailored for disadvantaged socioeconomic position: a systematic review and meta-regression. Lancet Public Health. 2019; 4(12):e628–e644. doi: 10.1016/S2468-2667(19)30220-8.
[41] Love-Koh J, Pennington B, Owen L, et al. How health inequalities accumulate and combine to affect treatment value: a distributional cost-effectiveness analysis of smoking cessation interventions. Soc Sci Med. 2020; 265:113339. doi: 10.1016/j.socscimed.2020.113339.
[42] NICE. Tobacco: preventing uptake, promoting quitting and treating dependence; 2025. Available from: https://www.nice.org.uk/guidance/ng209.
[43] UK CITaCR. Lung cancer incidence by deprivation 2020 [18 November 2025]; Calculated by the Cancer Intelligence Team at Cancer Research UK, April 2020. Based on method reported in National Cancer Intelligence Network Cancer by Deprivation in England Incidence, 1996–2010 Mortality, 1997–2011. Using cancer incidence data 2013–2017 (Public Health England) and population data 2013–2017 (Office for National Statistics) by Indices of Multiple Deprivation 2015 income domain quintile, cancer type, sex, and five–year age band.]. Available from: https://www.cancerresearchuk.org/health-professional/cancer-statistics/statistics-by-cancer-type/lung-cancer/incidence#lung-stats-incidence1.
[44] Kale D, Gilbert H, Sutton S. An exploration of the barriers to attendance at the english stop smoking services. Addict Behav Rep. 2019;9:005–005. doi: 10.1016/j.abrep.2018.10.005.

[45] Bolin K, Wilson K, Benhaddi H, et al. Cost-effectiveness of varenicline compared with nicotine patches for smoking cessation-results from four European countries. Eur J Public Health. 2009;19(6):650–654. doi: 10.1093/eurpub/ckp075.
[46] Howard P, Knight C, Boler A, et al. Cost-utility analysis of varenicline versus existing smoking cessation strategies using the BENESCO Simulation model: application to a population of US adult smokers. Pharmacoeconomics. 2008;26(6):497–511. doi: 10.2165/00019053-200826060-00004.
[47] Lutz MA, Lovato P, Cuesta G. Cost-effectiveness analysis of varenicline versus existing smoking cessation strategies in Central America and the Caribbean using the BENESCO model. Hosp Pract (1995). 2012;40(1):24–34. doi: 10.3810/hp.2012.02.945.
[48] Stanton H. The millennium development goals and tobacco control: an opportunity for global partnership. Tob Control. 2005;14(6):432–432. doi: 10.1136/tc.2005.014423.
[49] Leonardi-Bee J, Jere ML, Britton J. Exposure to parental and sibling smoking and the risk of smoking uptake in childhood and adolescence: a systematic review and meta-analysis. Thorax. 2011;66(10):847–855. doi: 10.1136/thx.2010.153379.
[50] Geneva WHO. Why is tobacco a public health priority?. Available from: http://www.who.int/tobacco/health_priority/en/index.html.
[51] Vangeli E, Stapleton J, Smit ES, et al. Predictors of attempts to stop smoking and their success in adult general population samples: a systematic review. Addiction. 2011;106(12):2110–2121. doi: 10.1111/j.1360-0443.2011.03565.x.
[52] Lewer D, McKee M, Gasparrini A, et al. Socioeconomic position and mortality risk of smoking: evidence from the English Longitudinal Study of Ageing (ELSA). Eur J Public Health. 2017;27(6):1068–1073. doi: 10.1093/eurpub/ckx059.
[53] Nixon SA. The coin model of privilege and critical allyship: implications for health. BMC Public Health. 2019;19(1):1637. doi: 10.1186/s12889-019-7884-9.